David Liu

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EDUCATION

University of Western Ontario — London, ON

B.E.Sc. in Software Engineering — *Sep 2022 – Apr 2028 (Expected)*

Awards: Among Us IRL, Best Use of Solana and Badge Hack ($2,500), Hack the North 2026; 2nd Place ($5,000), TELUS AI at the Edge of Innovation; Top 5, UofT Hackathon; Finalist, Western Engineering Competition

Involvements: AI Research Assistant, Tsinghua (2023); VP Education, Western Founders Network; Hack Western Mentor

Interests: Ignition Hacks Judge; Muay Thai coach; classical guitarist (Carnegie Hall)

PROFESSIONAL EXPERIENCE

J.D. Power — London, ON

Software Engineering Intern (16-month Co-op) — *May 2025 – Aug 2026*

- Built and maintained **Java** and **Spring Boot** services on **Kubernetes** for pricing and parts analysis across five Stellantis brands (Jeep, RAM, Dodge, Chrysler, Maserati), spanning **10+** repos
- Diagnosed **MY2026** spec and pricing failures across service repos and the data pipeline, using browser network traces to isolate the failing service; fixed the data mappings and filters and cut launch errors on the vehicle rollout
- Built a **Go** service and data pipeline that ingested, validated and loaded **20+** specs across five brands, automating spec intake; shipped through **GitLab** and **Jenkins CI/CD** and handled **load tests** and **incident response**
- Scanned **80+** repos with **TruffleHog**, remediated **2,000+** findings, and built a front-end dashboard to filter and triage vulnerabilities and trigger automated remediation; authored the **GitHub Copilot** research report that produced the scan findings

Tethos Association — London, ON

Founder and President — *May 2024 – Current*

- Solo-founded and scaled Western's largest student-run tech organization to **100+** developers and **150+** members across **2** chapters (Western, UBC), now a federally incorporated non-profit
- Delivered **20+** pro bono software projects for **World Vision**, the **Canadian Red Cross**, and the **London Children's Museum**, saving partners **$500,000+** in development costs
- Produced **GENESIS**, London's largest student demo day: **260+** attendees, **7** project demos, **8** sponsors, and **$12,000** in sponsorships

Modern Engineering — Delta, BC

Software & Robotics Engineering Intern — *May 2024 – Sep 2024*

- Programmed a **UR5** collaborative robot arm in **Python** over the **RTDE** interface: a **6**-program motion library spanning object retrieval, conveyor placement, part sanding, and **OpenCV** camera-guided stacking

PERSONAL PROJECTS

Tethos Island – Multiplayer Member Platform — May 2026 – Current

Next.js, TypeScript, Three.js, React Three Fiber, Colyseus, Supabase, PostgreSQL, Fly.io | GitHub | play.tethos.ca

- Built a browser multiplayer island for Tethos's **150+** members: a **Colyseus** server at **20** updates/sec with a **120-300 ms** interpolation buffer, a **50**-unit interest radius, and **30**-player shards that cap at **40**; a 40-player shard costs **3%** more than a 24-player one.
- Wrote the core world algorithms in **TypeScript**: **A*** with a binary heap and string pulling for **12** resident NPC schedules, a **47**-tile autotiler with rotation canonicalization, an **O(n)** distance transform for shorelines, and a fixed-step **120 Hz** character controller; merged terrain and cliff meshes to cut draw calls from **697** to **252**.
- Modeled the platform in **PostgreSQL** with **94** migrations and **131** tables behind **136** API routes; the server-authoritative economy runs through rate-limited RPCs and closed an item-duplication exploit, with **2,887** tests across **358** files.
- Built **5** LLM NPCs on **DeepSeek V4.1 Flash** with per-player memory: the server extracts structured facts from each chat, merges them into a memory set, and reloads them before the next conversation, with rate limits and a monthly interaction cap.

Wren – Multi-Tenant Nonprofit Platform — 2026 – Current

TypeScript, Next.js, PostgreSQL, Supabase, Trigger.dev, AWS, Terraform | Founder and engineer

- Built a multi-tenant SaaS on **Supabase Postgres**: row-level security on every table, isolation tests that prove cross-org reads fail, and a commit gate that blocks service-role use and unscoped views
- Routed every LLM call through one package with job-based model selection and a per-org **cost ledger**
- Built a spreadsheet intake with **PHI/SIN** guards, storage upload, commit and undo, backed by **10** migrations and **23** tests
- Built idempotent **Trigger.dev** jobs, **Sentry**/**PostHog** observability, and an **AWS** S3/SQS/ECS stack in **Terraform**

TECHNICAL SKILLS

Languages: Python, Java, C++, Go, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Express, Spring Boot, FastAPI, Flask, React Three Fiber, GSAP, Framer Motion, Tailwind

AI & Data: OpenAI, Anthropic & DeepSeek APIs, RAG, OpenCV, PyTorch, PostgreSQL, Supabase, MySQL, MongoDB

Systems & Tools: ESP32/PlatformIO, Docker, Kubernetes, AWS, Linux, Git, GitLab CI, Jenkins, GitHub Actions, REST APIs